HOSPITAL READMISSIONS

LEARNING TEAM 4

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BACKGROUND + CONTEXT

Hospital Quality Metrics:

Focus on readmission rates and expenses

Key Conditions:

Pneumonia, heart attack, knee replacement, and heart failure

Financial Impact:

High readmission rates lead to financial penalties and lower hospital ratings

Challenge:

Identifying factors that increase readmissions and prioritizing hospitals for intervention

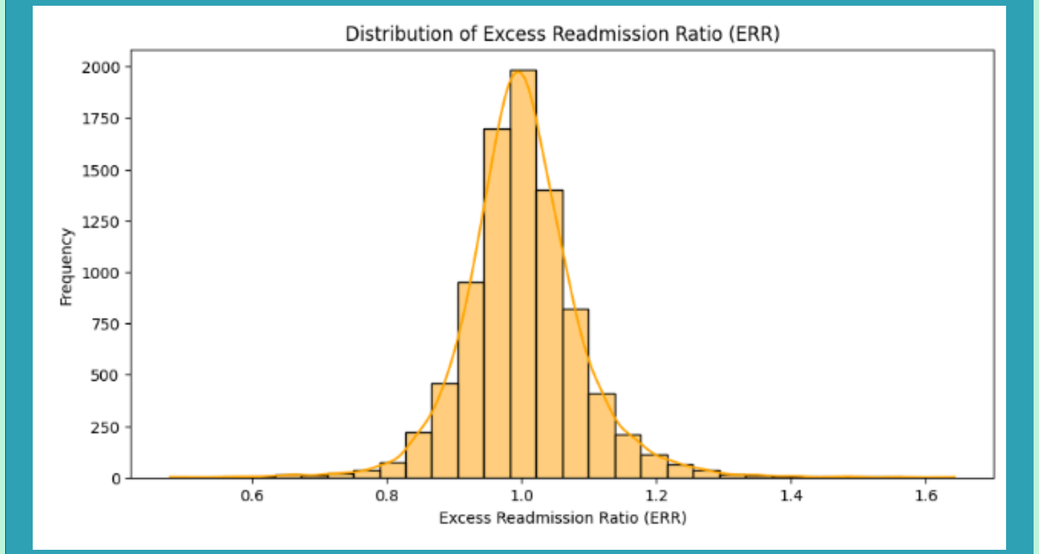
Data Insights:

Using analytics to find patterns and improve hospital effectiveness

PROBLEM STATEMENT

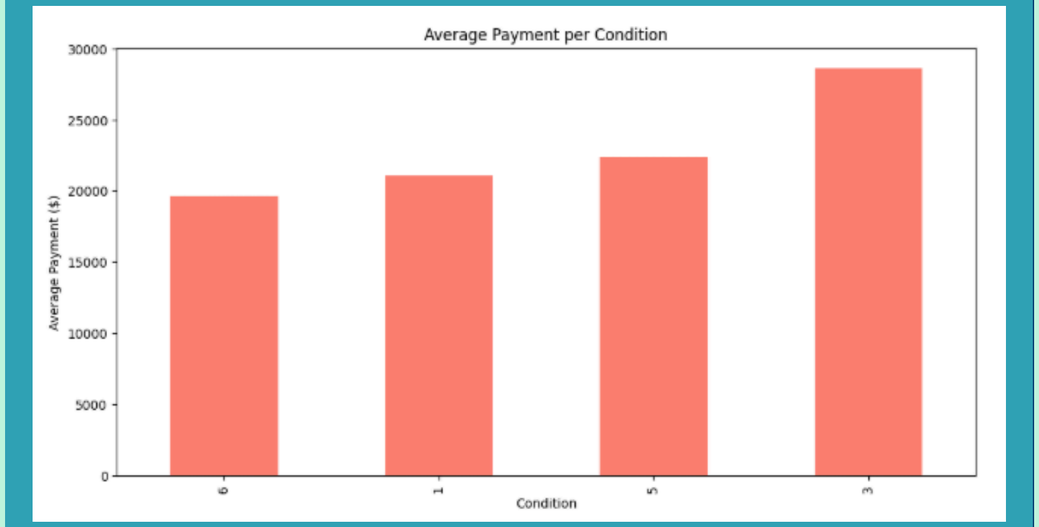
Given the excess *readmission ratio*, *payment*, and *type of medical condition*, can a reliable model be made to determine which factors lead to the highest increase in excess readmissions, and further, which hospitals are in the most need of interventions to cut down readmission rates in the United States?

EXPLORATORY DATA ANALYSIS



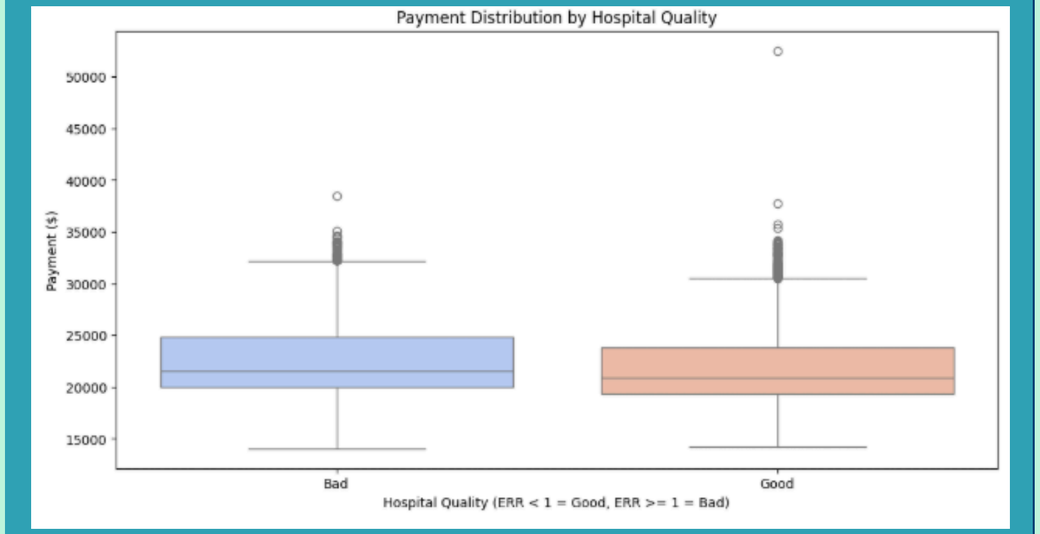
- Normally distributed with a mean of about 1
- ERR has a 0.4 ERR range between 2 STD
- Due to the large sample size
  - As sample size grows it represents the population more

- order from left to right:
  hf,p,h/kr, ha
- We assume this is because of the severity and extended cost of surgery, resulting in patients opting out of some care



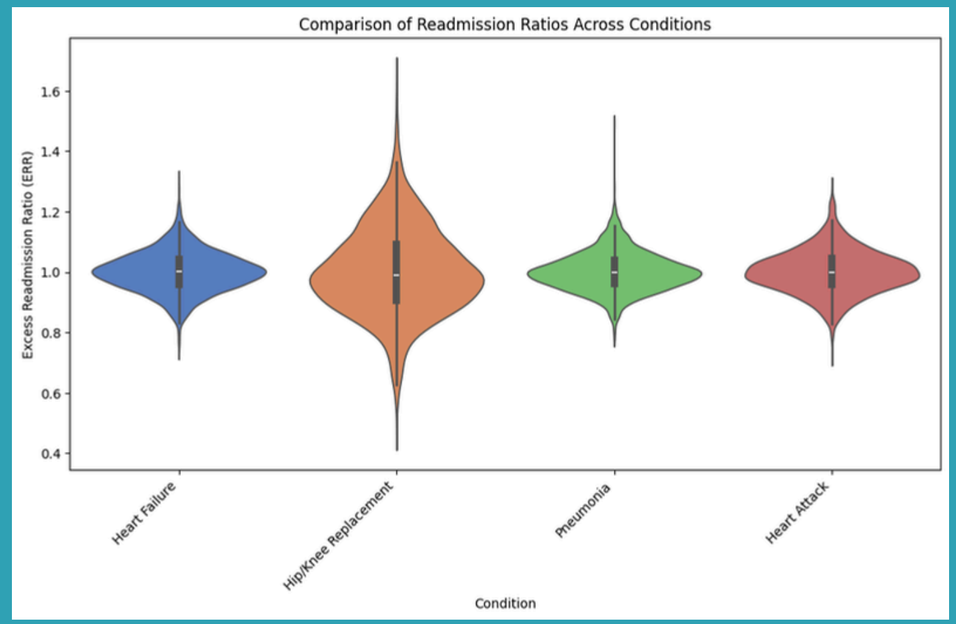
{1: "Pneumonia(p)", 3: "Heart Attack(hf)", 5: "Hip/Knee Replacement(h/kr)", 6: "Heart Failure(hf)"}

EXPLORATORY DATA ANALYSIS



- Majority of patients are in for H/KR
- ERR is very similar for all surgeries with the exception of H/KR.
- Pneumonia and H/KR has an unexpectedly high upper quartile.

- There is a very small difference between good and bad hospitals
- There are similar upper and lower quartiles
- Payments are not able to predict bad hospitals on its own



METHODOLOGY: CLASSIFICATION

7 Models were used

Gradient Boosted Trees (GBT)
Decision Tree,
Logistic Regression,
One-vs-Rest (OvR) Linear SVC,
Random Forest
Naïve Bayes.

- Used *K-fold cross-validation (k=3)* & *paramGrid search* to optimize hyperparameters for each model.
- 80% training, 20% testing, evaluating the predicting accuracy.

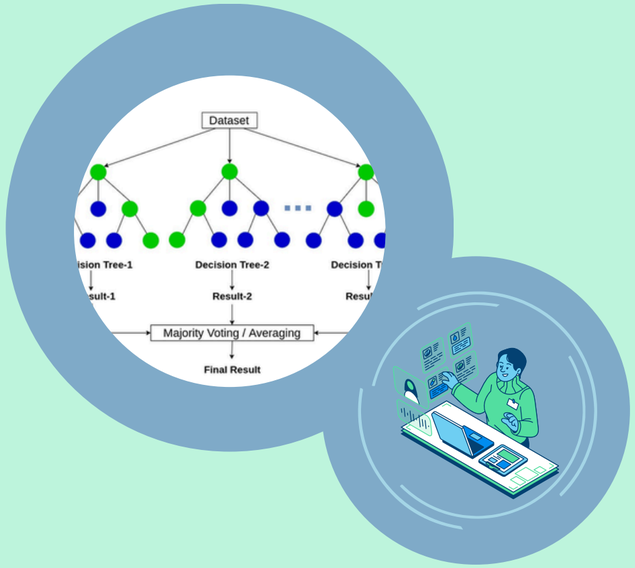
Model Implementation

Scaling & Feature Processing

Model Training & tuning

Categorical Features → encoded
Numerical Features → scaled
Pipeline → for consistency and scalibility

FINDINGS: CLASSIFICATION



MODEL PERFORMANCE

- Random forest achieved the highest AUC (0.632), slightly outperformed Logistic Regression (0.630).
- Decision Tree identified most bad hospitals (TP = 694).

KEY INSIGHTS

- Random Forest / Logistic Regression: Avoid mis-flagging good hospitals
- Decision Tree: Identify as many bad hospitals as possible

METHODOLOGY 2: CLUSTERING

Does payment influence readmission rates?

Cluster 1	Cluster 2	Cluster 3	Cluster 4
High Readmission Rate	Low Readmission Rate	High Readmission Rate	Low Readmission Rate
High Payment	High Payment	Low Payment	Low Payment

METHODOLOGY 2: CLUSTERING

Cluster 1

High Readmission Rate

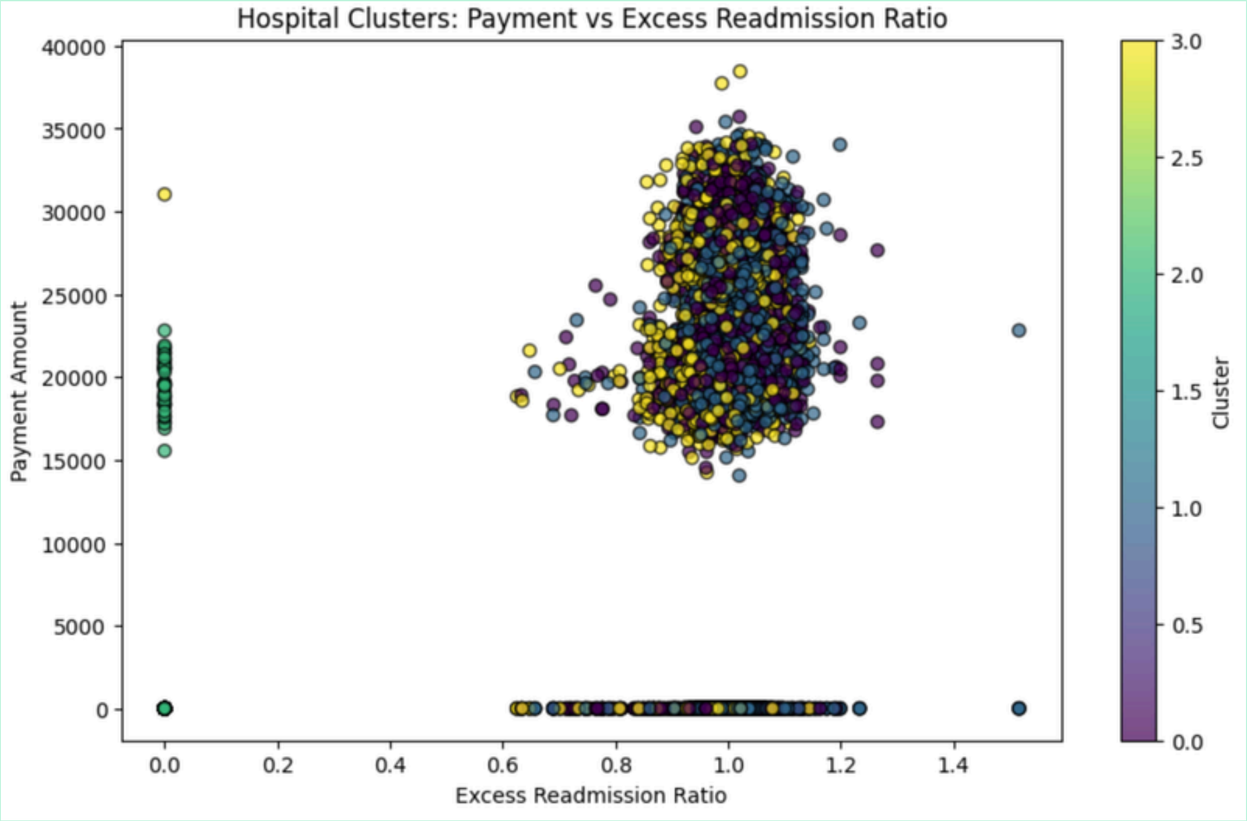
High Payment

Payment Measure Name

(Health Condition)

Payment State

FINDINGS FROM CLUSTERING



Most clusters have similar payments and average readmission ratios

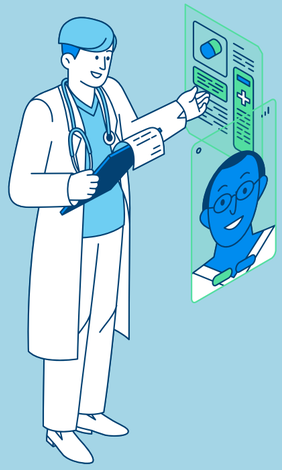
Few outliers, with similar traits

Similar health condition and state trends across different clusters

RECOMMENDATIONS

01 Use predictive models like Decision trees to spot potential poor outcomes early and avoid unnecessary treatments.

02 Compare hospitals to find cost inefficiencies and learn from the best-performing ones.



03 Enhance patient care by understanding their demographics, health conditions, and follow-up needs.

04 Use early warning systems to identify rising readmission risks and intervene before outcomes worsen.

CHALLENGES + MITIGATION

CLASSFICATION

HYPERPARAMETER TUNING

Exhaustive search of all combinations for multiple models is time and resource comsuming

PARAMGRID SEARCH

ParamGrid Search is feasible for datasets with a relatively small range

CLUSTERING

HANDLING “NOT AVAILABLE”

There are over 8,000 “not available” values under “Payment” column

REPLACE WITH 0 OR MEAN

We used 0 for now,
A better data collection needs to be implemented

VISUALIZATION

ERR DISTRIBUTION

Excess Readmission Rate is almost perfectly normally distributed, the data would be skewed if biased data is introduced

LARGE DATASET

Our dataset have a large sample size, theoretically this will result in normal distribution

THANK YOU!

QUESTIONS?

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